

Complexity of Finite Maps and Energy–Entropy Curves

Exchange rates, and a comparison that runs in circles

Artem Vorozhtsov
 artem.vorozhtsov@gmail.com

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Abstract

How many copies of one map between finite sets is a copy of another worth? Keep only the list of fiber sizes. One map is then built from another with a single call, after relabelling inputs injectively and outputs bijectively, exactly when the sorted lists fit entry by entry, and running maps side by side multiplies the lists. Trading many copies at once turns this into an exchange rate, which we compute: it is the smallest value of $\log Z_g / \log Z_f$, where $Z_a(\beta) = \sum_i a_i^\beta$ interpolates between counting the fibers and taking the largest one. Substituting $\epsilon_i = -\log a_i$ turns Z_a into a partition function and the answer into a picture — the largest factor by which the region under one energy–entropy curve can be enlarged, from the origin, and still fit under the other — with the map itself supplying the origin that the picture needs. The two directions give different rates, and the comparison they induce runs in circles: $\{5, 3\}$, $\{3, 1, 1\}$ and $\{6, 1\}$ each beat the next. So no number is hiding behind that comparison; what plays the part of complexity is the curve, taken up to scale. The same picture compares any two systems with finitely many energy levels, and for ultrarelativistic sectors the enlargement is exact — between the photon and electron–positron gases the factor is $7/4$.

Keywords: exchange rate; asymptotic conversion; fiber signature; partition function; Gibbs curve; energy–entropy region; resource theory.

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1 Introduction

Take two lists of positive integers,

$$\{2, 2\} \quad \text{and} \quad \{3, 1\}.$$

Say that one list *fits* into another when, after sorting both decreasingly, every entry of the first is at most the entry facing it in the second. Neither of these two fits into the other: $\{2, 2\}$ fails because its second entry 2 exceeds 1, and $\{3, 1\}$ fails because its first entry 3 exceeds 2.

Lists can be multiplied. The product $a \otimes b$ is the list of all products $a_i b_j$, keeping repetitions. So

$$\{3, 1\}^{\otimes 2} = \{9, 3, 3, 1\}, \quad \{2, 2\}^{\otimes 4} = \underbrace{\{16, \dots, 16\}}_{16},$$

and now the first *does* fit into the second. A single $\{3, 1\}$ cannot be built from a single $\{2, 2\}$, yet two copies of $\{3, 1\}$ can be built from four copies of $\{2, 2\}$. Three copies cannot, and neither can two copies from three, so this trade is the best available at these sizes.

This paper is about the best trade available in the long run. Write $k_{\max}(n)$ for the largest k such that $f^{\otimes k}$ fits into $g^{\otimes n}$, and let

$$C(g \rightarrow f) = \lim_{n \rightarrow \infty} \frac{k_{\max}(n)}{n}$$

be the number of copies of f that one copy of g is worth. For the two lists above,

$$C(\{2, 2\} \rightarrow \{3, 1\}) = \frac{\log 2}{\log 3} = 0.6309\dots, \quad C(\{3, 1\} \rightarrow \{2, 2\}) = 0.9653\dots$$

Two features are already visible. The two numbers are not reciprocal, so the trade cannot be undone: something is lost in either direction. And they are not equal, so it is tempting to call $\{3, 1\}$ the more complex of the two, since it buys more than it costs.

Such lists are not arbitrary. For a map $f : X \rightarrow Y$ from a finite set onto another, record how many inputs land on each output; the resulting list of fiber sizes is the *signature* of f . Fitting then means that f can be computed with a single call to g , after relabelling inputs injectively and outputs bijectively, and multiplying lists means running maps side by side. Section 2 makes this precise, and it is the last piece of set-up the paper needs.

Three things follow.

The first is that the limit above has a closed form. Attach to a list the weighted counts

$$Z_a(\beta) = \sum_i a_i^\beta, \quad \beta \geq 0,$$

which interpolate between the number of entries, at $\beta = 0$, and the largest entry, as $\beta \rightarrow \infty$. Then $C(g \rightarrow f)$ is the smallest value of $\log Z_g / \log Z_f$ over all β . Counting entries and taking the largest entry are the two obvious ways to compare lists; the content of the formula is that the truth lies at whichever β is worst, and that this is sometimes neither endpoint.

The second is that the formula is a picture. Substituting $\epsilon_i = -\log a_i$ turns Z_a into a partition function and β into an inverse temperature, so every list becomes a curve in the plane whose axes are mean energy and entropy. The exchange rate then has nothing to do with arithmetic: it is the largest factor by which the region under one curve can be enlarged, from the origin, and still slide under the other. Where the two curves touch is a temperature, and it is the bottleneck of the trade.

The third is that “more complex” is not a ranking. Comparing the two directed rates gives a relation that runs in circles: $\{5, 3\}$, $\{3, 1, 1\}$ and $\{6, 1\}$ each beat the next and are beaten by the one after. This is worth more than its curiosity value. Had the comparison come from some numerical complexity φ , the quantity $\log(C(a \rightarrow b)/C(b \rightarrow a))$ would be the difference $\varphi(b) - \varphi(a)$ and would sum to zero around every closed loop; around this one it does not, as Section 9 computes. No scalar lies behind the comparison, and what plays the part of complexity here is the whole curve, taken up to scale.

Nothing in the second description mentions integers, and this is what makes the picture worth having. Any system with a finite list of energy levels has such a curve, and two systems can be compared the same way. When the curves happen to be exact enlargements of one another the trade is reversible: the cleanest instance is a gas of photons against a gas of electrons and positrons in the early universe, where the factor is exactly $7/4$. Sections 10–13 collect this and other examples, from spins in a magnetic field to crystal-field levels to an Ising ring.

The paper is meant to be read in order, and everything up to Section 13 is elementary. Connections to resource theories, to Weihrauch reducibility and to Strassen’s asymptotic spectrum are deferred to Section 14, where they can be stated quickly and skipped freely.

2 Signatures and implementation

Let $f : X \rightarrow Y$ be onto and enumerate its output as $Y = \{y_1, \dots, y_r\}$. The *fiber* over y_i is the inverse image

$$f^{-1}(y_i) = \{x \in X : f(x) = y_i\}, \quad n_i = |f^{-1}(y_i)|.$$

Order the fiber sizes decreasingly and define the *signature* of f by

$$\sigma(f) = (n_1, \dots, n_r), \quad n_1 \geq \dots \geq n_r \geq 1, \quad r = |Y|.$$

No other information about the map will be used, and we henceforth use the same letter for a map and its signature.

Definition 1 (Implemented by). Let $a = (a_1, \dots, a_r)$ and $b = (b_1, \dots, b_s)$ be decreasing signatures. We say that a is *implemented by* b , and write $a \preceq_{\text{impl}} b$, when $r \leq s$ and

$$a_i \leq b_i \quad (1 \leq i \leq r).$$

For maps, this means that there is an injective input relabeling h_{in} and a bijection from the used output fibers to the output of f such that

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}.$$

More precisely, put

$$X'_g = h_{\text{in}}(X_f), \quad Y'_g = g(X'_g).$$

The two inclusions on the right identify the parts of X_g and Y_g actually used by the implementation, and the following diagram commutes:

$$\begin{array}{ccccc} X_f & \xrightarrow[\cong]{h_{\text{in}}} & X'_g & \hookrightarrow & X_g \\ f \downarrow & & \downarrow g|_{X'_g} & & \downarrow g \\ Y_f & \xleftarrow[\cong]{h_{\text{out}}} & Y'_g & \hookrightarrow & Y_g \end{array}$$

Unused fibers of g may be discarded. Such an implementation exists precisely when each fiber of f injects into its assigned fiber of g , which is the componentwise condition in Definition 1.

Cartesian product multiplies fiber sizes:

$$a \otimes b = \text{sort}\{a_i b_j : 1 \leq i \leq r, 1 \leq j \leq s\}.$$

Multiplicity is retained, and $a^{\otimes k}$ denotes the k -fold product.

Disjoint union supplies a second operation:

$$f \sqcup g : X_f \sqcup X_g \longrightarrow Y_f \sqcup Y_g$$

has signature

$$a + b = \text{sort}(a_1, \dots, a_r, b_1, \dots, b_s).$$

Both operations respect implementation, and Cartesian product distributes over this sum:

$$a \preceq_{\text{impl}} b, c \preceq_{\text{impl}} d \implies a + c \preceq_{\text{impl}} b + d, \quad a \otimes c \preceq_{\text{impl}} b \otimes d, \quad a \otimes (b + c) = (a \otimes b) + (a \otimes c).$$

Here $+$ is a labelled alternative and $\otimes$ is parallel composition. Only the product enters the exchange rate; the sum is recorded because it is needed once, in Section 14.

Definition 2 (Exchange rate). For signatures f and g , set

$$k_{\max}^{g \rightarrow f}(n) = \max \left\{ k \geq 0 : f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n} \right\}.$$

The exchange rate is

$$C(g \rightarrow f) = \lim_{n \rightarrow \infty} \frac{k_{\max}^{g \rightarrow f}(n)}{n}.$$

Throughout, *the implementing resource is written first*: $C(g \rightarrow f)$ counts copies of the target f obtained per copy of the resource g , so the superscript $g \rightarrow f$ records the direction of the conversion. Larger values mean that g is the more useful resource.

The construction goes back to [1], where a relation of this kind was introduced informally and called a complexity relation, not only for maps but for other objects as well — convex plane figures, natural numbers and information channels among them. The present paper makes it precise for finite maps, computes the rate, and finds the convex figures again in Section 6.

Tensoring optimal implementations at block lengths n and m gives

$$k_{\max}(n + m) \geq k_{\max}(n) + k_{\max}(m).$$

Fekete's lemma now yields

$$\lim_{n \rightarrow \infty} \frac{k_{\max}(n)}{n} = \sup_{n \geq 1} \frac{k_{\max}(n)}{n},$$

with $+\infty$ allowed. The limit is the operational definition of $C(g \rightarrow f)$.

3 Power sums and Gibbs curves

For a signature $a = (a_i)$, define its power-sum partition function

$$Z_a(\beta) = \sum_i a_i^\beta, \quad \beta \geq 0,$$

where $\beta = 1/T$ is inverse temperature. The semiring operations become ordinary addition and multiplication:

$$Z_{a+b}(\beta) = Z_a(\beta) + Z_b(\beta), \quad Z_{a \otimes b}(\beta) = Z_a(\beta) Z_b(\beta).$$

The fibers of the k -fold Cartesian power $a^{\otimes k}$ are indexed by $(i_1, \dots, i_k)$ and have sizes $a_{i_1} \cdots a_{i_k}$. Therefore

$$\begin{aligned} Z_{a^{\otimes k}}(\beta) &= \sum_{i_1, \dots, i_k} (a_{i_1} \cdots a_{i_k})^\beta \\ &= \left(\sum_i a_i^\beta \right)^k = Z_a(\beta)^k, \quad \log Z_{a^{\otimes k}}(\beta) = k \log Z_a(\beta). \end{aligned} \tag{1}$$

Proposition 1. Let f and g each have at least two fibers and a fiber of size greater than one. Then

$$C(g \rightarrow f) = \inf_{\beta \in [0, \infty]} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \tag{2}$$

where the value at $\beta = \infty$ is

$$\frac{\log \max_j g_j}{\log \max_i f_i}.$$

The proof is deferred to Section 5.

Put

$$\epsilon_i = -\log a_i, \quad p_i(T) = \frac{e^{-\epsilon_i/T}}{\sum_j e^{-\epsilon_j/T}} = \frac{a_i^{1/T}}{Z_a(1/T)}.$$

The corresponding mean energy and entropy are

$$E_a(T) = \sum_i p_i(T) \epsilon_i, \quad H_a(T) = -\sum_i p_i(T) \log p_i(T).$$

They satisfy

$$\log Z_a(1/T) = H_a(T) - \frac{E_a(T)}{T}.$$

Since the Gibbs distribution of a Cartesian power is a product distribution,

$$H_{a^{\otimes k}}(T) = kH_a(T), \quad E_{a^{\otimes k}}(T) = kE_a(T),$$

and therefore

$$E_{a^{\otimes k}}(H) = kE_a(H/k),$$

so Cartesian powers act by homothety in the (E, H) -plane.

Let Γ_a denote the Gibbs curve $\{(E_a(T), H_a(T)) : 0 < T < \infty\}$, completed by its endpoint segments, and let $\mathcal{R}_a$ be the closed convex region between this boundary and the line $E = 0$. The quantity

$$\log Z_a(\beta) = H_a(T) - \beta E_a(T)$$

is the offset of the supporting line with normal $(-\beta, 1)$. Proposition 1 is therefore equivalent to

$$C(g \rightarrow f) = \sup\{c \geq 0 : c\mathcal{R}_f \subseteq \mathcal{R}_g\}.$$

Equivalently, $C(g \rightarrow f)^{-1}\Gamma_g$ is the smallest homothetic copy of Γ_g whose region contains Γ_f .

4 The pair $\{2, 2\}$ and $\{3, 1\}$

Let $g = \{2, 2\}$ and $f = \{3, 1\}$. All 2^n fibers of $g^{\otimes n}$ have the same size 2^n , so $f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n}$ asks only two things: that $f^{\otimes k}$ have no more fibers, and that its largest fiber not exceed 2^n . The first condition is $2^k \leq 2^n$, that is $k \leq n$; the second is $3^k \leq 2^n$, that is $k \leq n \log 2 / \log 3$. Since $\log 2 / \log 3 < 1$, the second is the binding one, and the count is exact:

$$k_{\max}(n) = \left\lfloor n \frac{\log 2}{\log 3} \right\rfloor, \quad C(\{2, 2\} \rightarrow \{3, 1\}) = \frac{\log 2}{\log 3} = 0.6309297536 \dots$$

The two trades computed by hand in the introduction are the cases $n = 3$ and $n = 4$, where the floor is 1 and 2. The reverse direction is limited at a finite temperature:

$$\begin{aligned} C(\{3, 1\} \rightarrow \{2, 2\}) &= \min_{\beta \geq 0} \frac{\log(3^\beta + 1)}{(\beta + 1) \log 2} \\ &= 0.9653844417 \dots, \quad \beta_* = 0.4036799030 \dots, \quad T_* = \beta_*^{-1} = 2.47721027 \dots \end{aligned}$$

Figure 1 displays both homotheties. Solid curves are unscaled; dotted curves are enlarged by the reciprocal directed rate. Thin endpoint segments complete the boundaries, and the fills show the regions $\mathcal{R}_a$. Because the two levels of $\{2, 2\}$ coincide, its Gibbs curve reduces to the point $(-\log 2, \log 2)$.

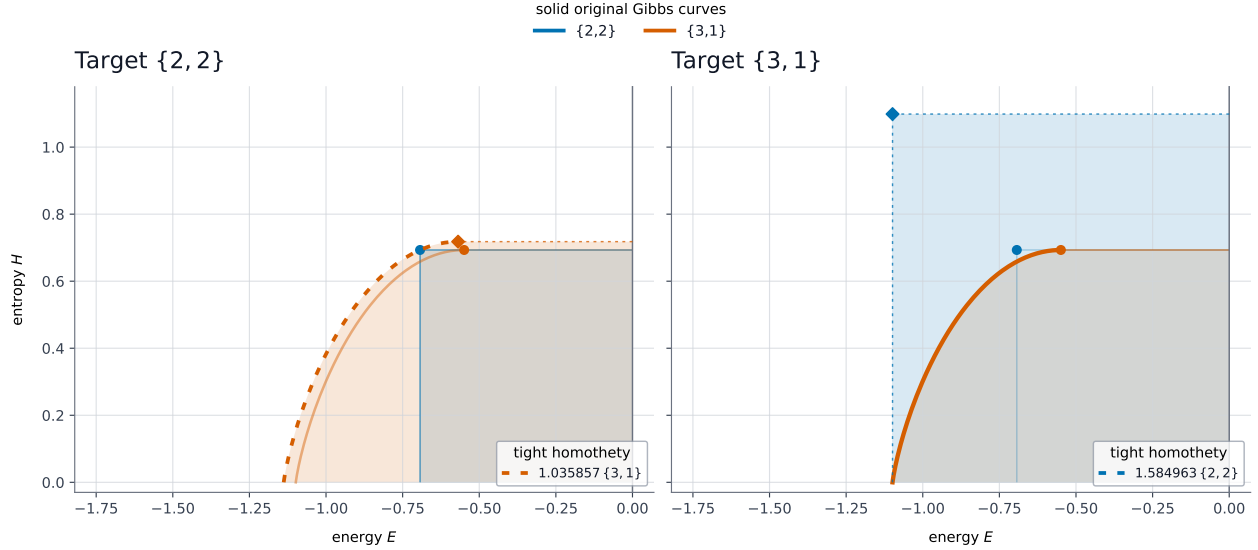


Figure 1: Exchange rates as planar dilations. In each panel, one energy–entropy region is scaled about the origin until it just contains the other. Thin segments at the $T = 0$ and $T = \infty$ endpoints close the regions; dotted boundaries denote the scaled copy, and translucent fills make the containment visible. The required scale factors are $1/C(\{3, 1\} \rightarrow \{2, 2\}) = 1.03585676\dots$ on the left and $1/C(\{2, 2\} \rightarrow \{3, 1\}) = \log 3 / \log 2$ on the right.

5 Proof of the rate formula

This section and the next are the technical ones. Nothing later depends on their details, so a reader who would rather go on to further examples can skip to Section 7.

Lemma 1 (Counting form of $\preccurlyeq_{\text{impl}}$). For signatures a and b ,

$$a \preccurlyeq_{\text{impl}} b \iff \#\{i : a_i \geq t\} \leq \#\{j : b_j \geq t\} \quad \text{for every } t \geq 1.$$

Proof. If $a \preccurlyeq_{\text{impl}} b$ then $r \leq s$ and $a_i \leq b_i$, so every $a_i \geq t$ forces $b_i \geq t$ and the left count is at most the right one. Conversely, apply the hypothesis at $t = a_i$: at least i entries of a are $\geq a_i$, hence so are at least i entries of b , and as b decreases this says $b_i \geq a_i$. Taking $t = 1$ gives $r \leq s$. $\square$

Proof of Proposition 1. Write $\varphi_a(\beta) = \log Z_a(\beta)$. Every $a_i \geq 1$, so Z_a is non-decreasing and $\varphi_a(\beta) \geq \varphi_a(0) = \log(\text{number of fibers}) \geq \log 2 > 0$. Hence $R(\beta) = \varphi_g(\beta)/\varphi_f(\beta)$ is continuous and positive on $[0, \infty)$. Moreover $\varphi_a(\beta) = \beta \log a_1 + \log m_a + o(1)$, where m_a is the number of fibers of maximal size, so R extends continuously to $[0, \infty]$ with the value stated at ∞ . Its infimum C_* is therefore attained and lies in $(0, \infty)$.

The rate is at most C_ .* Suppose $f^{\otimes k} \preccurlyeq_{\text{impl}} g^{\otimes n}$. For $\beta \geq 0$ the map $x \mapsto x^\beta$ is non-decreasing on $[1, \infty)$, so matching fibers one by one and discarding the unused ones gives $Z_f(\beta)^k \leq Z_g(\beta)^n$, that is $k/n \leq R(\beta)$. Matching the largest fibers gives $f_1^k \leq g_1^n$, which is $k/n \leq R(\infty)$. Hence $k_{\max}(n)/n \leq C_*$ for every n .

The rate is at least C_ .* Fix integers $k, n \geq 1$ with $k/n < C_*$. We show $f^{\otimes k\ell} \preccurlyeq_{\text{impl}} g^{\otimes n\ell}$ for all large ℓ . Put $u_i = \log f_i$ and $v_j = \log g_j$, and for $m \geq 1$, $x \in \mathbb{R}$ let

$$N_f(m, x) = \#\{\omega \in \{1, \dots, r\}^m : u_{\omega_1} + \dots + u_{\omega_m} \geq mx\}$$

count the fibers of $f^{\otimes m}$ of size at least e^{mx} , and define N_g likewise. A threshold t corresponds to $z = (\log t)/\ell$, so by Lemma 1 it is enough to prove

$$N_f(k\ell, z/k) \leq N_g(n\ell, z/n) \quad \text{for every } z \in \mathbb{R}. \quad (3)$$

Introduce

$$\psi_a(x) = \inf_{\beta \geq 0} (\varphi_a(\beta) - \beta x),$$

which is the Gibbs curve of this section in disguise: where the infimum is attained, $\varphi_a(\beta) - \beta x = H_a(T)$ with $x = -E_a(T)$, so ψ_a is entropy as a function of energy.

Upper bound. For every $\beta \geq 0$,

$$N_f(m, x) \leq \sum_{\omega} e^{\beta(u_{\omega_1} + \dots + u_{\omega_m} - mx)} = e^{m(\varphi_f(\beta) - \beta x)},$$

whence $N_f(m, x) \leq e^{m\psi_f(x)}$, with $e^{-\infty} = 0$.

A uniform gap. Since $k/n < C_* \leq R(\beta)$ and $\varphi_f(\beta) > 0$, the function $D(\beta) = n\varphi_g(\beta) - k\varphi_f(\beta)$ is positive on $[0, \infty)$; and $k/n < R(\infty)$ gives $k \log f_1 < n \log g_1$, so $D(\beta) \rightarrow +\infty$. A positive continuous function tending to infinity has a positive infimum, so $\delta := \inf_{\beta \geq 0} D(\beta) > 0$. As $k\psi_f(z/k) = \inf_{\beta \geq 0} (k\varphi_f(\beta) - \beta z)$, and similarly for g , this yields

$$n\psi_g(z/n) \geq k\psi_f(z/k) + \delta \quad \text{for every } z \in \mathbb{R}. \quad (4)$$

Lower bound, and where the endpoint is used. If $z > k \log f_1$ the left side of (3) vanishes and there is nothing to prove. So assume $z \leq k \log f_1$; then

$$\frac{z}{n} \leq \frac{k}{n} \log f_1 < \log g_1, \quad \eta := \log g_1 - \frac{k}{n} \log f_1 > 0,$$

and η does not depend on z . This is the only place where the endpoint $\beta = \infty$ enters, and it is what makes the counting uniform: whenever $f^{\otimes k\ell}$ has any fiber above the threshold at all, the matching threshold for $g^{\otimes n\ell}$ stays a fixed distance below the largest fiber of g . Immediately below that largest fiber the counts grow polynomially rather than exponentially, and there an exponential-rate estimate would decide nothing.

Fix m large enough that $2s \log g_1/m \leq \eta/2$, and let $L = L(g, \eta)$ bound the minimising β in $\psi_g(x)$ over $x \leq \log g_1 - \eta/2$; on that range ψ_g is Lipschitz with the same constant. We claim

$$N_g(m, x) \geq e^{m\psi_g(x) - c \log(m+1)} \quad (x \leq \log g_1 - \eta) \quad (5)$$

for a constant $c = c(g, \eta)$. Indeed, put $x' = x + 2s \log g_1/m$, let β attain the infimum for $\psi_g(x')$, and let $p_j = g_j^\beta / Z_g(\beta)$. Round p to a type P with denominator m , so $\|P - p\|_1 \leq 2s/m$. Every word of type P satisfies $m^{-1} \sum_a v_{\omega_a} = \sum_j P_j v_j \geq x' - 2s \log g_1/m = x$, the type class has at least $(m+1)^{-s} e^{mH(P)}$ elements [7], and $H(P) \geq H(p) - O(s \log m/m)$ while $H(p) = \psi_g(x') \geq \psi_g(x) - 2Ls \log g_1/m$.

Conclusion. Combining the upper bound with (4) and (5) at $m = n\ell$,

$$\log N_g(n\ell, z/n) - \log N_f(k\ell, z/k) \geq \ell\delta - c \log(n\ell + 1),$$

positive for every $\ell \geq \ell_0$, where ℓ_0 depends only on f, g, k, n and not on z . So (3) holds for all z once $\ell \geq \ell_0$, giving $f^{\otimes k\ell} \preceq_{\text{impl}} g^{\otimes n\ell}$ and $k_{\max}(n\ell) \geq k\ell$. Hence $C(g \rightarrow f) \geq k/n$, and letting k/n increase to C_* completes the proof. The counting is the one used for exact asymptotic majorization; compare [8]. $\square$

6 Products, areas, and negative temperatures

Three properties of the region $\mathcal{R}_a$ are worth recording. The first says where its boundary stops, and why.

Remark 1 (Why the curve is not continued past $\beta = 0$). Nothing forbids negative β , and the curve does continue there: past its peak at $\beta = 0$ it turns and descends, returning towards the origin as $\beta \rightarrow -\infty$. For $\{3, 1\}$ the peak is $(-0.549, 0.693)$, and $\beta = -3$ gives $(-0.039, 0.154)$. Systems with a bounded spectrum really do reach such states — nuclear spins and cold atoms in optical lattices — so it is worth saying why they play no part here.

The reason is that $Z_a(\beta)$ is order preserving only for $\beta \geq 0$. For $\beta < 0$ the map $x \mapsto x^\beta$ reverses order, and fitting no longer forces an inequality between power sums: $\{1\} \preceq_{\text{impl}} \{2\}$, yet at $\beta = -\frac{1}{2}$ one has $Z_{\{1\}} = 1 > Z_{\{2\}} = 0.707\dots$. Negative β would therefore contribute constraints that are simply false. Moreover $\log Z_a(\beta)$ can turn negative there — for $\{2, 2\}$ already at $\beta = -2$ — so the inequality $k/n \leq \log Z_g / \log Z_f$ would change direction and the formula would lose its meaning.

Closing the region instead by the horizontal segment at $H = \log r$ is the right thing to do, and it reads plainly: a state with inverted populations is never the better resource, because one can always waste energy instead and keep the same entropy.

Products have a geometric meaning as well, though it is only the standard Legendre duality in other words.

Remark 2 (Product is Minkowski sum). This is immediate from the duality between $S(E)$ and $\log Z$ recalled above. That display says $\log Z_a(\beta)$ is the support function of $\mathcal{R}_a$ in the direction $(-\beta, 1)$; support functions add under Minkowski addition, and $\log Z$ adds under $\otimes$. Hence

$$\mathcal{R}_{a \otimes b} = \mathcal{R}_a \oplus \mathcal{R}_b = \{x + y : x \in \mathcal{R}_a, y \in \mathcal{R}_b\}.$$

For the directions not of the form $(-\beta, 1)$, write $u = (u_1, u_2)$: if $u_1, u_2 > 0$ the maximum of $\langle x, u \rangle$ sits at the corner $(0, \log r)$ and equals $u_2 \log Z_a(0)$; if $u_2 \leq 0$ it has $H = 0$ and equals 0 for $u_1 \geq 0$ and $|u_1| \log a_1$ otherwise. Both are again additive under $\otimes$.

So the two operations of this paper are the two operations of planar convex geometry: $\otimes$ is Minkowski addition, and Cartesian powers are dilations, $\mathcal{R}_{a^{\otimes k}} = k \mathcal{R}_a$.

Areas therefore grow quadratically, which gives a crude but instant bound on the exchange rate.

Proposition 2 (Area bound). Let A_a be the area of $\mathcal{R}_a$. Then $A_{a^{\otimes k}} = k^2 A_a$ and

$$C(g \rightarrow f) \leq \sqrt{A_g / A_f}.$$

Proof. The first claim is the dilation of Remark 2. For the second, $c \mathcal{R}_f \subseteq \mathcal{R}_g$ gives $c^2 A_f \leq A_g$. $\square$

The bound is honest but loose: for the pair of Section 4 it returns 0.856 against the true 0.631, and 1.168 against 0.965. It cannot be sharp, because by Remark 4 the exact rate is controlled by the multiplicative family $Z_a(\beta)$, whereas the area is quadratic and so is not one of them. By the Brunn–Minkowski inequality $\sqrt{A}$ is superadditive under $\otimes$, and the mixed area $V(\mathcal{R}_a, \mathcal{R}_b)$ in $A_{a \otimes b} = A_a + 2V(\mathcal{R}_a, \mathcal{R}_b) + A_b$ is a joint invariant of two signatures that the present theory does not otherwise see. We do not pursue it here.

7 A two-sided finite-temperature tangency

Both directed rates may be attained away from the endpoints. Consider

$$a = \{6, 5, 2, 1\}, \quad b = \{6, 4, 3, 2\}.$$

The two signatures have equal length and equal largest fiber, so both endpoint ratios are one:

$$\frac{\log Z_a(0)}{\log Z_b(0)} = 1, \quad \lim_{\beta \rightarrow \infty} \frac{\log Z_a(\beta)}{\log Z_b(\beta)} = \frac{\log 6}{\log 6} = 1.$$

The interior minima are:

Direction	C	T_*	$\beta_* = 1/T_*$
$C(a \rightarrow b)$	0.9664896992	1.7970865612	0.5564562229
$C(b \rightarrow a)$	0.9786567233	0.2459449336	4.0659508025

If

$$R(\beta) = \frac{\log Z_g(\beta)}{\log Z_f(\beta)}$$

has an interior minimum at β_* , then $R'(\beta_*) = 0$ and $\log Z = H - \beta E$ give

$$E_g(T_*) = C(g \rightarrow f)E_f(T_*), \quad H_g(T_*) = C(g \rightarrow f)H_f(T_*).$$

The target curve and the scaled source curve therefore meet at the same point. Their slopes also agree, since

$$\frac{dH}{dE} = \frac{1}{T_*} = \beta_*,$$

and this slope is invariant under homothety. The two contacts in Figure 2 are genuine tangencies.

8 A cyclic strict comparison

Definition 3. For signatures a, b , define

$$a \prec b \iff C(a \rightarrow b) < C(b \rightarrow a).$$

In this terminology, b is more complex than a when conversion from b to a is more efficient than conversion in the opposite direction. The relation is not transitive.

Set

$$f_1 = \{5, 3\}, \quad f_2 = \{3, 1, 1\}, \quad f_3 = \{6, 1\}.$$

Equation (2) gives the comparisons in Table 1.

Table 1: Directed rates producing a strict cycle.

Pair	Forward rate	Reverse rate	Conclusion
f_1, f_2	$C(f_1 \rightarrow f_2) = 0.630929754$	$C(f_2 \rightarrow f_1) = 0.670211928$	$f_1 \prec f_2$
f_2, f_3	$C(f_2 \rightarrow f_3) = 0.613147193$	$C(f_3 \rightarrow f_2) = 0.630929754$	$f_2 \prec f_3$
f_3, f_1	$C(f_3 \rightarrow f_1) = 0.897931851$	$C(f_1 \rightarrow f_3) = 0.898244402$	$f_3 \prec f_1$

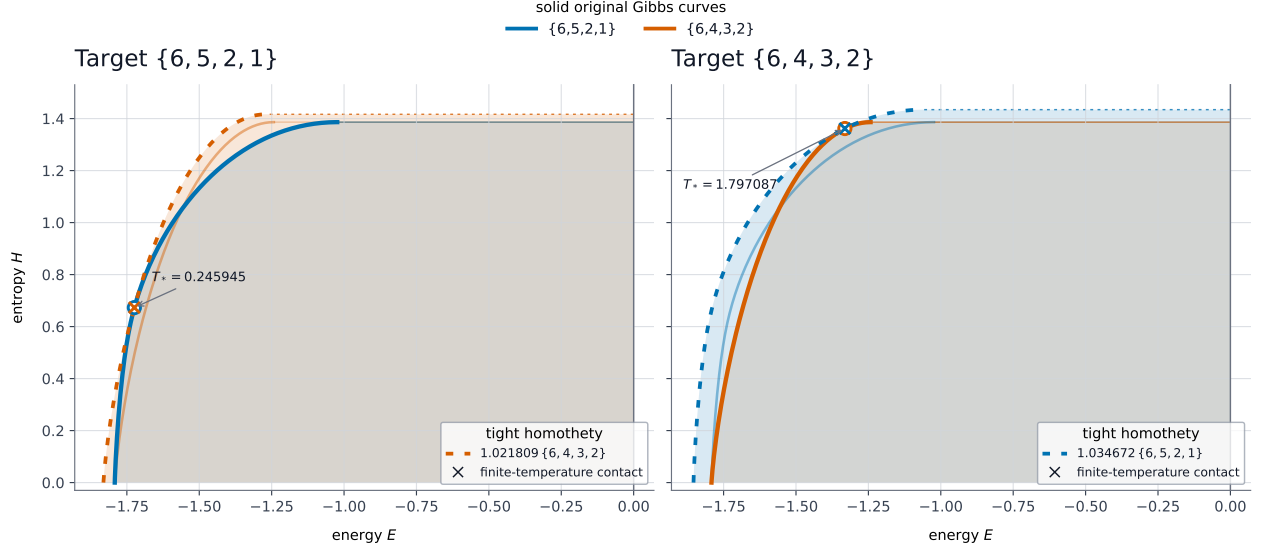


Figure 2: Two-sided finite-temperature tangency. The left panel scales b by $1/C(b \rightarrow a)$ to embrace a ; the right panel scales a by $1/C(a \rightarrow b)$ to embrace b . Crosses mark the interior contact points. Thin dotted horizontal lines close the scaled regions at their $T = \infty$ endpoints; translucent fills extend to $E = 0$.

The three rows give

$$f_1 \prec f_2, \quad f_2 \prec f_3, \quad f_3 \prec f_1,$$

and Appendix C proves them, four of the six rates being exactly $\log 2 / \log 3$, $\log 3 / \log 6$ or $\log 5 / \log 6$. Hence $\prec$ is not a strict order. Figure 3 shows the three comparisons geometrically. In the panel with target f_i , the dotted curve for f_j is $C(f_j \rightarrow f_i)^{-1} \Gamma_{f_j}$.

9 A thermodynamic analogy and its limits

Associate signatures $A = (a_i)$ and $B = (b_j)$ with two idealized particle species whose internal energy levels are

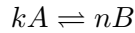
$$\epsilon_i^A = -\log a_i, \quad \epsilon_j^B = -\log b_j.$$

Then $A^{\otimes k}$ and $B^{\otimes n}$ represent k noninteracting particles of type A and n noninteracting particles of type B , respectively. If their scaled Gibbs curves are tangent at interior points,

$$\frac{dH_A}{dE_A} = \frac{dH_B}{dE_B} = \frac{1}{T}.$$

The tangent Gibbs states therefore have the same temperature.

This identification is kinematic, not dynamical: $\preceq_{\text{impl}}$ records an allowed combinatorial conversion, not a reaction mechanism. For the formal stoichiometry



write $r = k/n$. In the asymptotic closure,

$$B^{\otimes n} \rightarrow A^{\otimes k} \iff r \leq C(B \rightarrow A), \tag{6}$$

$$A^{\otimes k} \rightarrow B^{\otimes n} \iff r \geq \frac{1}{C(A \rightarrow B)}. \tag{7}$$

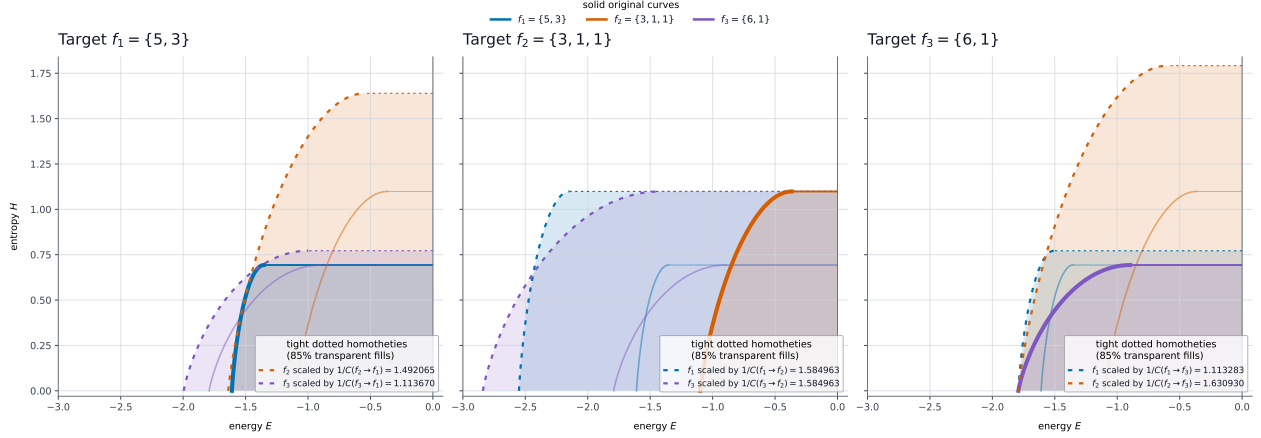


Figure 3: The cyclic example. Solid and dotted boundaries are original and scaled, respectively. Horizontal $T = \infty$ closures and 85%-transparent fills extending to $E = 0$ display target containment.

Strict inequalities guarantee the corresponding direction for sufficiently large block sizes; an exact boundary ratio need not be attained at finite size. Since

$$C(A \rightarrow B)C(B \rightarrow A) \leq 1,$$

the thresholds leave the interval

$$C(B \rightarrow A) \leq r \leq \frac{1}{C(A \rightarrow B)}.$$

For ratios in its interior, neither direction need realize the prescribed stoichiometry. Coincident endpoints mean asymptotic reversibility; otherwise the interval measures irreversibility.

Physical equilibrium requires more. At fixed temperature and pressure the reaction affinity must vanish:

$$\mathcal{A} = k\mu_A - n\mu_B = 0,$$

where μ_A and μ_B are chemical potentials. Neither chemical potentials nor reaction kinetics occur in our definition. Equations (6)–(7) therefore describe allowed conversions, not spontaneous reaction directions.

The strict cycle admits a formal comparison with a driven reaction network. For $a \xrightarrow{s} b$, set

$$\alpha(a \rightarrow b) = \log \frac{C(a \rightarrow b)}{C(b \rightarrow a)} > 0.$$

For the cycle of Section 8,

$$f_1 \xrightarrow{s} f_3 \xrightarrow{s} f_2 \xrightarrow{s} f_1,$$

the sum around the cycle is

$$\begin{aligned} \mathcal{A}_{\text{cyc}} &= \alpha(f_1 \rightarrow f_3) + \alpha(f_3 \rightarrow f_2) + \alpha(f_2 \rightarrow f_1) \\ &= \log \frac{0.898244402}{0.897931851} + \log \frac{0.630929754}{0.613147193} + \log \frac{0.670211928}{0.630929754} \\ &= 0.089337, \quad e^{\mathcal{A}_{\text{cyc}}} = 1.09345. \end{aligned}$$

If α is read as a logarithmic forward–reverse bias, the orientation matches the reaction network

$$A \rightleftharpoons B, \quad B \rightleftharpoons C, \quad C \rightleftharpoons A.$$

It cannot represent circulation in a closed equilibrium gas. Reaction affinity is the negative derivative of Gibbs energy with respect to reaction extent [10]; detailed balance forces the affinity around a closed cycle to vanish. Nonzero cycle affinity instead belongs to an open system maintained by reservoirs. Such networks are studied in [11, 12]; a stochastic three-species example appears in [13].

The analogy stops at this structural resemblance. The rates $C(a \rightarrow b)$ are not kinetic constants, and the three minima may occur at different temperatures. Without kinetics, conservation laws, or external work accounting, the strict cycle predicts neither a current nor an oscillation.

A fixed reaction energy can be included as an offset. For

$$kA \rightleftharpoons nB + q$$

shift every level of B by q/n :

$$\tilde{\epsilon}_{B,j} = \epsilon_{B,j} + \frac{q}{n}.$$

Then

$$\tilde{E}_B(H) = E_B(H) + \frac{q}{n}, \quad \tilde{Z}_B(1/T) = e^{-q/(nT)} Z_B(1/T),$$

while $H_B(T)$ and $dH_B/dE = 1/T$ are unchanged. The n -particle region is therefore translated horizontally by q . This accounts for a fixed offset carried by a work-storage degree of freedom; heat transfer to a bath cannot be represented by a horizontal shift alone.

The model omits interactions, quantum statistics, translational and volume degrees of freedom, mixing entropy, conservation laws, kinetics, activation barriers, and reservoirs. Its energies are dimensionless ($k_B = 1$), and relative offsets between particle types must be fixed whenever particle number can change. These omissions delimit the analogy.

The supporting function

$$\Psi_A(T) = \log Z_A(1/T) = -\frac{F_A(T)}{T}$$

is the Massieu potential. Proposition 1 becomes

$$C(A \rightarrow B) = \inf_{T>0} \frac{\Psi_A(T)}{\Psi_B(T)}$$

and asks for the largest c such that $\Psi_A(T) \geq c\Psi_B(T)$ for every T . An interior minimizer is the bottleneck temperature. As discussed in Section 2, our comparison uses the whole Gibbs boundary of a system rather than the states of a fixed one [16].

An exact physical homothety occurs for ultrarelativistic equilibrium sectors. If g_X is the effective number of internal degrees of freedom, then in natural units and at fixed volume

$$E_X(T) = \frac{\pi^2}{30} g_X V T^4, \quad H_X(T) = \frac{2\pi^2}{45} g_X V T^3.$$

Therefore

$$E_A(T) = \frac{g_A}{g_B} E_B(T), \quad H_A(T) = \frac{g_A}{g_B} H_B(T)$$

at every temperature, so the two boundaries are homothetic with factor g_A/g_B . For photons $g_\gamma = 2$; for an ultrarelativistic electron–positron sector $g_{e^\pm} = (7/8) \times 4 = 7/2$. The factor is therefore $7/4$, as in Figure 4. In the early universe, $e^- + e^+ \rightleftharpoons 2\gamma$ maintained equilibrium until the temperature fell below the electron mass and annihilation transferred entropy to the photon bath [18].

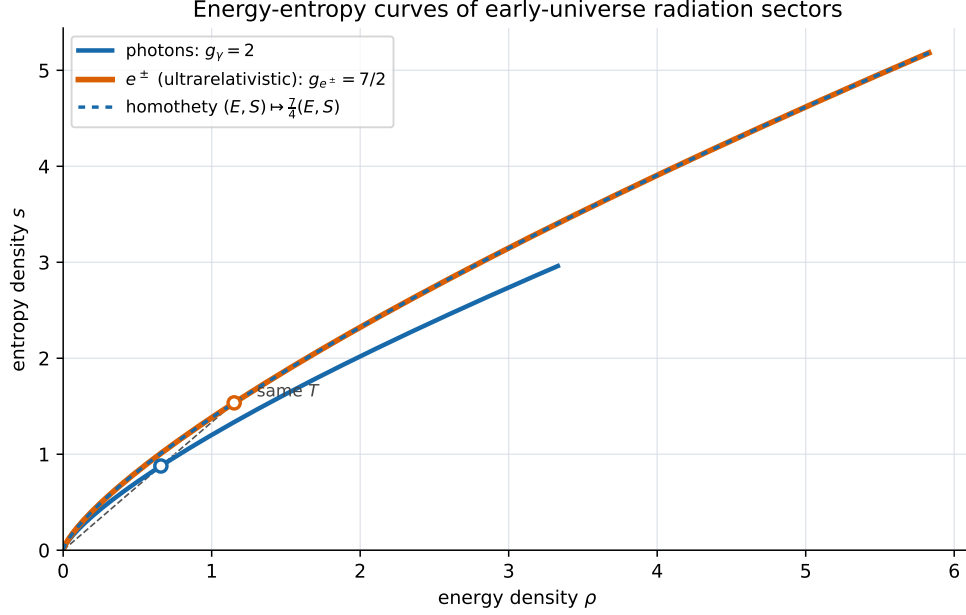


Figure 4: Exact homothety in the ultrarelativistic limit. At fixed temperature, the electron–positron sector has $7/4$ times the energy and entropy densities of the photon sector; the rescaled photon curve therefore coincides with it.

10 Gibbs systems with a finite spectrum

Nothing in Section 3 used integrality. Let a system have finitely many energy levels ϵ_i with degeneracies g_i , and put

$$Z(\beta) = \sum_i g_i e^{-\beta \epsilon_i}.$$

Independent composition multiplies partition functions, exactly as $\otimes$ does, so

$$C(A \rightarrow B) = \inf_{\beta \in [0, \infty]} \frac{\log Z_A(\beta)}{\log Z_B(\beta)}$$

is defined verbatim. Signatures are the special case in which every $e^{-\epsilon_i}$ is a positive integer and $g_i = 1$; repeated entries reproduce degeneracies, so $\{2, 2\}$ is a single doubly degenerate level.

Two invariance properties delimit what this measures.

Remark 3 (Scale but not shift). C is unchanged by a global rescaling $\epsilon_i \mapsto \lambda \epsilon_i$ with $\lambda > 0$, because β absorbs λ . It is *not* invariant under a shift $\epsilon_i \mapsto \epsilon_i + c$, since the homothety is centred at the origin. What the construction measures is therefore a property of the pair $(\Gamma_a, \text{origin})$, not of the curve Γ_a alone: two observers who place the zero of energy differently will disagree about C , and as the origin recedes every rate tends to 1. In the finite-map model the origin is supplied canonically — a fiber of size one carries no multiplicity, so $a_i = 1$ corresponds to $\epsilon_i = 0$. For a physical spectrum it has to be nominated instead. In the examples below we use the *top gauge*, in which the highest level is placed at $\epsilon = 0$; this is the choice that makes all weights $e^{-\epsilon_i} \geq 1$, as for signatures.

Proposition 3 (The two endpoints). Let A and B have finite spectra, dimensions $D = \sum_i g_i$ and depths $W = -\min_i \epsilon_i$ below the chosen origin. Then

$$\frac{\log Z_A}{\log Z_B} \Big|_{\beta=0} = \frac{\log D_A}{\log D_B}, \quad \lim_{\beta \rightarrow \infty} \frac{\log Z_A(\beta)}{\log Z_B(\beta)} = \frac{W_A}{W_B}.$$

Proof. At $\beta = 0$ one has $Z = D$. As $\beta \rightarrow \infty$, $\log Z(\beta) = \beta W + \log g_{\min} + o(1)$, where $g_{\min}$ is the degeneracy of the lowest level; the ratio tends to W_A/W_B . $\square$

So the two obvious ways to compare systems — how many states they have, and how wide their spectra are — are exactly the two endpoints, and an interior minimiser means that neither alone is the bottleneck. For signatures the two endpoints are $\log r$ and $\log a_1$.

11 Exact homotheties

Two regions are homothetic precisely when the Massieu potentials are proportional,

$$\Psi_A(T) = c \Psi_B(T) \text{ for all } T \iff Z_A = Z_B^c,$$

and then $C(A \rightarrow B) = c$, $C(B \rightarrow A) = 1/c$, so $C(A \rightarrow B)C(B \rightarrow A) = 1$: homothety is exactly asymptotic reversibility.

Extensivity. For N non-interacting copies $Z_N = Z_1^N$, so the N -particle region is the single-particle region scaled by N . The homothety of Cartesian powers in Section 3 is therefore nothing but thermodynamic extensivity, and $C(A \rightarrow B)$ is a generalised, not necessarily integer, particle count.

Massless bosonic sectors in three dimensions. A massless bosonic sector with g internal degrees of freedom and propagation speed v , in a fixed volume V , has

$$E = \frac{\pi^2}{30} \frac{g}{v^3} VT^4, \quad H = \frac{2\pi^2}{45} \frac{g}{v^3} VT^3, \quad \Psi = \frac{\pi^2}{90} \frac{g}{v^3} VT^3.$$

Any two such sectors are therefore exactly homothetic, with

$$c = \frac{g_A/v_A^3}{g_B/v_B^3}. \quad (8)$$

This one formula generates a family of exact homotheties, of which the electron–positron example of Figure 4 is the first row:

sector A	sector B	c
$e^\pm$ ($g = 7/2$)	photons ($g = 2$)	$7/4$
three neutrino species ($g = 21/4$)	photons ($g = 2$)	$21/8$
Debye phonons ($g = 3$, speed $\bar{c}_s$)	photons ($g = 2$, speed c)	$\frac{3}{2}(c/\bar{c}_s)^3$

The last row is the most extreme: with $\bar{c}_s = 5 \text{ km s}^{-1}$ the factor is $3 \cdot 10^{14}$. A cold crystal and a photon gas have *identical* energy–entropy shape and differ only by a dilation of fourteen orders of magnitude. Here $\bar{c}_s$ is the effective sound speed defined by $3/\bar{c}_s^3 = 1/c_L^3 + 2/c_T^3$.

Each homothety holds within the scaling regime that produces it: the $e^\pm$ row requires $T \gg m_e$, and the Debye row requires $T \ll \Theta_D$. Outside these windows the proportionality of Ψ fails, and the comparison becomes one of the contact problems of the next section.

12 Finite-temperature contact

When the two potentials are not proportional, the scaled region touches its target and the contact locates a bottleneck temperature. All values below are computed in the top gauge of Remark 3; Δ denotes the level spacing and drops out of C by Remark 3.

Relativistic plasmas across thresholds. For a plasma in equilibrium at a single temperature, $\Psi \propto g_*(T)VT^3$, so (2) collapses to

$$C(A \rightarrow B) = \min_T \frac{g_*^A(T)}{g_*^B(T)},$$

the minimum over temperature of the ratio of effective degrees of freedom. Comparing the Standard Model plasma with a pure photon gas over the window $1 \text{ MeV} < T < 300 \text{ GeV}$, where g_* runs from 10.75 to 106.75 [17],

$$C(\text{SM} \rightarrow \gamma) = \frac{10.75}{2} = 5.375, \quad C(\gamma \rightarrow \text{SM}) = \frac{2}{106.75} = 0.018735,$$

with product 0.1007. The two contacts sit at opposite ends of the window: the bottleneck is the epoch at which the two plasmas differ most. The restriction to a temperature window is essential — after neutrino decoupling the plasma no longer has a single temperature.

Dimension against bandwidth. A spin-1/2 and a spin-1 in the same field have spectra $\{0, -\Delta\}$ and $\{0, -\Delta, -2\Delta\}$. By Proposition 3 the endpoints are $\log 3 / \log 2$ and 2, and both rates are attained at endpoints:

$$C(\text{spin-1} \rightarrow \text{spin-}\frac{1}{2}) = \frac{\log 3}{\log 2} = 1.584963, \quad C(\text{spin-}\frac{1}{2} \rightarrow \text{spin-1}) = \frac{1}{2},$$

with product $\log 3 / \log 4 = 0.792481$. The spin-1 is dimension-limited as a resource and the spin-1/2 is bandwidth-limited. Two qubits with gaps Δ and 2Δ give the extreme case: the wider one implements the narrower at rate 1, the narrower implements the wider only at rate 1/2.

Crystal-field schemes. For Ce^{3+} in a cubic field the $J = 5/2$ multiplet splits into a Γ_7 doublet and a Γ_8 quartet. Compare the two orderings: scheme A has the doublet lowest, scheme B the quartet. Both have dimension 6 and the same total splitting, so *both endpoints equal one*, yet

$$C(A \rightarrow B) = 0.849525 \text{ at } T_* = 0.481814 \Delta, \quad C(B \rightarrow A) = 1.$$

The scheme with the degenerate level lowest is the strictly better resource. With $\Delta = \log 3$ these are the integer signatures $\{3, 3, 3, 3, 1, 1\}$ and $\{3, 3, 1, 1, 1, 1\}$, so this example lies in both halves of the paper at once. The $J = 4$ cubic scheme of Pr^{3+} , with irreducible representations of dimensions 1, 2, 3, 3, gives 0.827934 at $T_* = 0.300661 \Delta$ between its two orderings.

Interaction as complexity. A four-spin Ising ring at coupling J and four free spins in a field $h = J$ have the same dimension 16 and the same bandwidth $8J$, so again both endpoints are one. Nevertheless

$$C(\text{Ising} \rightarrow \text{free}) = 1, \quad C(\text{free} \rightarrow \text{Ising}) = 0.959587 \text{ at } T_* = 0.666839 J,$$

so the interacting ring is strictly the more complex of the two, and the obstruction sits at the coupling scale. At matched dimension and matched bandwidth, a state count cannot see this difference; the Gibbs region can.

Table 2: Directed rates for the physical examples. Contacts marked $T = \infty$ are dimension-limited, $T = 0$ bandwidth-limited, and a finite T_* is an interior tangency.

A	B	$C(A \rightarrow B)$	$C(B \rightarrow A)$	product	contact
<i>exact homotheties (reversible)</i>					
$e^\pm$	photons	1.75	0.571429	1	all T
3ν	photons	2.625	0.380952	1	all T
Debye phonons	photons	$3 \cdot 10^{14}$	$3 \cdot 10^{-15}$	1	all T
N copies	one copy	N	$1/N$	1	all T
<i>finite-temperature contact (irreversible)</i>					
SM plasma	photons	5.375	0.018735	0.100703	window ends
spin-1	spin-1/2	1.584963	0.500000	0.792481	$T = \infty, T = 0$
qubit 2Δ	qubit Δ	1.000000	0.500000	0.500000	$T = \infty, T = 0$
Ce ³⁺ inverted	Ce ³⁺ normal	1.000000	0.849525	0.849525	$T = 0, T_* = 0.4818\Delta$
Pr ³⁺ inverted	Pr ³⁺ normal	1.000000	0.827934	0.827934	$T = 0, T_* = 0.3007\Delta$
Ising ring	free spins	1.000000	0.959587	0.959587	$T = 0, T_* = 0.6668J$

13 Summary of the examples, and limits

Table 2 collects the comparisons. Reversible rows are exactly the homothetic pairs of Section 11.

Three limitations delimit the construction.

First, unbounded spectra with different scaling exponents give nothing. Ferromagnetic magnons have $\epsilon \propto k^2$ and hence $\Psi \propto T^{3/2}$, against T^3 for photons; the ratio runs over all of $(0, \infty)$ and both directed rates vanish. A non-trivial comparison requires matching asymptotics at both ends of the temperature range, which is automatic for finite spectra and otherwise has to be imposed — for instance by restricting the infimum to a physical window $T \in [T_{\min}, T_{\max}]$, as was done for the plasma above.

Second, the construction assumes concavity of $H(E)$. Self-gravitating systems violate it: a Schwarzschild black hole has $H \propto E^2$ and negative heat capacity, so the supporting-line duality behind (2) does not apply.

Third, by Remark 3 the origin is part of the data. The construction is canonical without further input exactly when the boundary passes through the origin, that is when $E \rightarrow 0$ and $H \rightarrow 0$ as $T \rightarrow 0$. This holds for the scale-free sectors of Section 11, which is why those examples require no convention, and fails for gapped finite systems, where a gauge must be fixed.

14 Relations to other work

Adjoining the empty signature as a zero, signatures form an ordered commutative semiring with multiplicative unit $\{1\}$. This places the construction among semiring resource theories: implementation is the free conversion, $+$ is disjoint choice, and $\otimes$ combines independent resources. General accounts of resource theories and their ordered algebraic structures are given in [2, 3].

The one-shot factorization

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}$$

has the form of strong Weihrauch reduction: one call to g , preceded and followed by fixed processors [4]. Unlike the computable processors in Weihrauch theory, ours are injective or bijective on the sets actually used. This restriction preserves the fiber capacities.

The asymptotic construction is closer to tensor restriction. Tensors encode bilinear tasks, linear maps implement restrictions, and tensor product combines copies. Strassen’s asymptotic spectrum characterizes large tensor powers by multiplicative monotones [5]. The relative exponent

$$\omega(t, s) = \lim_{n \rightarrow \infty} \frac{1}{n} \min\{m : t^{\otimes m} \text{ restricts to } s^{\otimes n}\}$$

measures resource cost per target copy; it is used, for example, to quantify irreversibility in matrix multiplication [6]. Its convention is reciprocal to ours after reversing the conversion. The resulting formula is also analogous to exact asymptotic majorization, where the rate is a minimum of Rényi-entropy ratios [8].

Remark 4 (The power sums are exactly the monotones). A *monotone* for a resource theory is a quantity attached to each resource that cannot increase under the free conversions — entropy is the familiar example. Strassen’s theory of the asymptotic spectrum [5] singles out those monotones φ that are compatible with the whole algebraic structure: additive under $+$, multiplicative under $\otimes$, monotone under $\preceq_{\text{impl}}$, and equal to 1 on the unit $\{1\}$. For signatures these are precisely the power sums. Indeed, every signature is a sum of singletons, $\{a_1, \dots, a_r\} = \{a_1\} + \dots + \{a_r\}$, so φ is determined by the numbers $t_n = \varphi(\{n\})$. Since $\{n\} \otimes \{m\} = \{nm\}$, the sequence t_n is completely multiplicative, and since $\{n\} \preceq_{\text{impl}} \{m\}$ exactly when $n \leq m$, it is non-decreasing. A non-decreasing completely multiplicative sequence with $t_1 = 1$ is $t_n = n^\beta$ for a unique $\beta \geq 0$; monotonicity applied to $2^p \leq n^q$ gives $t_2^p \leq t_n^q$, and letting p/q approach $\log n / \log 2$ from both sides forces $\log t_n / \log n = \log t_2 / \log 2$. Hence $\varphi(a) = \sum_i a_i^\beta = Z_a(\beta)$, and the family $\{Z_a(\beta)\}_{\beta \geq 0}$ is a complete set of monotones in the sense that (2) determines every exchange rate. The endpoint $\beta = \infty$, that is $a \mapsto \max_i a_i$, is not itself such a monotone, because the maximum is not additive under $+$; it enters (2) only as a limit, and by continuity the infimum over $[0, \infty)$ is unchanged by adjoining it. General accounts of monotones for ordered algebraic resource structures are given in [3].

The energy–entropy plane is itself an established tool in quantum thermodynamics, and the closest existing framework to the one used here. Sparaciari, Oppenheim, and Fritz construct a resource theory without a background temperature in which states are classified by their average energy and entropy, and conversion rates are read off a two-dimensional convex diagram [16]. The difference is what the diagram collects. Theirs is the set of pairs (E, H) realised by the *states* of one fixed Hamiltonian, and the free operations act within that system; ours is the Gibbs boundary of a system, and the comparison is between *different* spectra, with scaling by a homothety about the origin in place of a background bath. Relatedly, thermomajorization compares two states of a fixed system by asking whether one concave curve dominates another [14], and the conditions for conversion are governed by a family of Rényi free energies indexed by a single parameter [15], in close formal analogy with the family $Z_a(\beta)$ of Section 3. What appears to be new here is the use of homothetic containment as a *directed* comparison between systems, and the resulting failure of transitivity in Section 6.

Channel simulation provides a third comparison: an encoder and decoder use copies of one channel to reproduce another. With common-randomness assistance, the first-order rate for point-to-point discrete memoryless channels is the ratio of their capacities [9]. Appendix A recovers the corresponding logarithmic cardinality ratio when both processors in the finite-map model are left unrestricted.

15 Conclusion

A fiber signature is a coarse invariant of a finite map, but Cartesian powers already give it a nontrivial asymptotic conversion theory. The operational rate, the power-sum formula, and containment of Gibbs regions are equivalent descriptions of that theory. The directed rates cannot be compressed into a global ranking, as the explicit three-cycle shows. Extending Gibbs-region containment beyond finite signatures remains plausible, but it requires an operational model distinct from chemical reaction dynamics.

Code and data availability

Every rate, table, and figure in this paper is reproducible from the reference implementation at github.com/avorozhtsov/fn_complexity, which contains the exchange rate solver, the exact finite-power verifier $k_{\max}$, the cached matrix of directed rates, the shell-screening code behind Appendix B, and the scripts that generate the figures.

References

- [1] A. V. Vorozhtsov, “Algebraicheskie metody opredeleniya slozhnosti” [Algebraic methods for defining complexity], Preprint No. 17, Keldysh Institute of Applied Mathematics, Moscow (2001), in Russian. library.keldysh.ru/preprint.asp?id=2001-17.
- [2] B. Coecke, T. Fritz, and R. W. Spekkens, “A mathematical theory of resources,” *Information and Computation* **250**, 59–86 (2016). doi:10.1016/j.ic.2016.02.008.
- [3] T. Fritz, “Resource convertibility and ordered commutative monoids,” *Math. Struct. Comput. Sci.* **27**, 850–938 (2017). doi:10.1017/S0960129515000444.
- [4] V. Brattka, G. Gherardi, and A. Pauly, “Weihrach complexity in computable analysis,” in *Handbook of Computability and Complexity in Analysis*, pp. 367–417 (Springer, Cham, 2021). doi:10.1007/978-3-030-59234-9_11.
- [5] V. Strassen, “The asymptotic spectrum of tensors,” *J. Reine Angew. Math.* **384**, 102–152 (1988). doi:10.1515/crll.1988.384.102.
- [6] M. Christandl, P. Vrana, and J. Zuiddam, “Barriers for fast matrix multiplication from irreversibility,” *Theory Comput.* **17**(2), 1–32 (2021). doi:10.4086/toc.2021.v017a002.
- [7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, Hoboken, 2006), Chapter 11.
- [8] A. Kjærulff Jensen, “Asymptotic majorization of finite probability distributions,” *IEEE Trans. Inf. Theory* **65**, 8131–8139 (2019). doi:10.1109/TIT.2019.2922627.
- [9] M. X. Cao, N. Ramakrishnan, M. Berta, and M. Tomamichel, “Channel simulation: finite blocklengths and broadcast channels,” *IEEE Trans. Inf. Theory* **70**, 6780–6808 (2024). doi:10.1109/TIT.2024.3445998; arXiv:2212.11666.
- [10] International Union of Pure and Applied Chemistry, “Affinity of reaction,” *Compendium of Chemical Terminology (the Gold Book)*, 5th ed., online version 5.0.0 (2025). doi:10.1351/goldbook.A00178.

- [11] M. Polettini and M. Esposito, “Irreversible thermodynamics of open chemical networks I: emergent cycles and broken conservation laws,” *J. Chem. Phys.* **141**, 024117 (2014). doi:10.1063/1.4886396.
- [12] R. Rao and M. Esposito, “Nonequilibrium thermodynamics of chemical reaction networks: wisdom from stochastic thermodynamics,” *Phys. Rev. X* **6**, 041064 (2016). doi:10.1103/PhysRevX.6.041064.
- [13] T. Schmiedl and U. Seifert, “Stochastic thermodynamics of chemical reaction networks,” *J. Chem. Phys.* **126**, 044101 (2007). doi:10.1063/1.2428297.
- [14] M. Horodecki and J. Oppenheim, “Fundamental limitations for quantum thermodynamics,” *Nature Communications* **4**, 2059 (2013). doi:10.1038/ncomms3059.
- [15] F. Brandão, M. Horodecki, N. Ng, J. Oppenheim, and S. Wehner, “The second laws of quantum thermodynamics,” *Proc. Natl. Acad. Sci. USA* **112**(11), 3275–3279 (2015). doi:10.1073/pnas.1411728112.
- [16] C. Sparaciari, J. Oppenheim, and T. Fritz, “A resource theory for work and heat,” *Phys. Rev. A* **96**, 052112 (2017). doi:10.1103/PhysRevA.96.052112.
- [17] E. W. Kolb and M. S. Turner, *The Early Universe*, Frontiers in Physics **69** (Addison-Wesley, Redwood City, 1990).
- [18] L. C. Thomas, T. Dezen, E. B. Grohs, and C. T. Kishimoto, “Electron-positron annihilation freeze-out in the early universe,” *Phys. Rev. D* **101**, 063507 (2020). doi:10.1103/PhysRevD.101.063507.

A Unrestricted input and output maps

Suppose that the maps h_{in} and h_{out} in the definition of implementation are unrestricted. Then b is implementable by a if and only if

$$|Y_b| \leq |Y_a|.$$

An arbitrary h_{in} may collapse each fiber of b onto one chosen input of a , so only the number of available output fibers remains. A deterministic h_{out} cannot increase that number.

For Cartesian powers the condition is therefore

$$|Y_b|^k \leq |Y_a|^n, \quad k_{\max}(n) = \left\lfloor n \frac{\log |Y_a|}{\log |Y_b|} \right\rfloor.$$

Hence, for $|Y_a|, |Y_b| > 1$,

$$C(a \rightarrow b) = \frac{\log |Y_a|}{\log |Y_b|}.$$

The same ratio appears in assisted point-to-point channel interconversion [9]: a surjective deterministic channel with output set Y has capacity $\log |Y|$. Unrestricted processors retain only this capacity; the processors of Definition 1 also retain the individual fiber sizes.

$C(a \rightarrow b)C(b \rightarrow a) = 1$, so the comparison from Definition 3 reduces to

$$a \prec b \iff |Y_a| < |Y_b|.$$

This is a total preorder, and quotienting by equal output cardinality gives a linear order. In partition-function language only $Z_a(0) = |Y_a|$ remains. If $|Y_b| = 1$, then $C(a \rightarrow b) = +\infty$; if $|Y_a| = 1 < |Y_b|$, then $C(a \rightarrow b) = 0$.

B The first 69 non-special signatures

We omit the exceptional infinite families $\{N\}$ and $\{1, \dots, 1\}$. To enumerate every other signature without imposing a fixed length or entry bound, put

$$\mathcal{S}_B = \{a : |a| \geq 2, a_1 > 1, a_1 + 2|a| \leq B\}.$$

Every non-special signature eventually belongs to $\mathcal{S}_B$. For each B , the strongly connected components of the strict comparison graph are contracted and topologically ordered by the deterministic rule implemented in the analysis code. At the stated screening precision, the first 69 entries agree for $B = 18, 19, 20$. Table 3 records this stable prefix; its rates were recomputed from the high-accuracy cache.

Because $\prec$ is not transitive, this list is not a linear extension of all pairwise comparisons. A row for x in Table 4 records each earlier a for which

$$C(x \rightarrow a) < C(a \rightarrow x)$$

despite the displayed order $a < x$.

Table 3: The stabilized first 69 non-special signatures in the deterministic condensation-DAG order.

n	signature	n	signature	n	signature
1	$\{2, 1\}$	24	$\{6, 6\}$	47	$\{9, 8\}$
2	$\{2, 2\}$	25	$\{7, 1\}$	48	$\{9, 9\}$
3	$\{3, 1\}$	26	$\{7, 2\}$	49	$\{10, 1\}$
4	$\{2, 1, 1\}$	27	$\{7, 3\}$	50	$\{10, 2\}$
5	$\{3, 2\}$	28	$\{7, 4\}$	51	$\{10, 3\}$
6	$\{2, 2, 1\}$	29	$\{7, 5\}$	52	$\{10, 4\}$
7	$\{3, 3\}$	30	$\{7, 6\}$	53	$\{10, 5\}$
8	$\{2, 1, 1, 1\}$	31	$\{7, 7\}$	54	$\{10, 6\}$
9	$\{2, 2, 2\}$	32	$\{8, 1\}$	55	$\{10, 7\}$
10	$\{4, 1\}$	33	$\{8, 2\}$	56	$\{10, 8\}$
11	$\{4, 2\}$	34	$\{8, 3\}$	57	$\{10, 9\}$
12	$\{4, 3\}$	35	$\{8, 4\}$	58	$\{10, 10\}$
13	$\{4, 4\}$	36	$\{8, 5\}$	59	$\{11, 1\}$
14	$\{5, 1\}$	37	$\{8, 6\}$	60	$\{11, 2\}$
15	$\{5, 2\}$	38	$\{8, 7\}$	61	$\{11, 3\}$
16	$\{5, 3\}$	39	$\{8, 8\}$	62	$\{11, 4\}$
17	$\{5, 4\}$	40	$\{9, 1\}$	63	$\{11, 5\}$
18	$\{5, 5\}$	41	$\{9, 2\}$	64	$\{11, 6\}$
19	$\{6, 1\}$	42	$\{9, 3\}$	65	$\{11, 7\}$
20	$\{6, 2\}$	43	$\{9, 4\}$	66	$\{11, 8\}$
21	$\{6, 3\}$	44	$\{9, 5\}$	67	$\{11, 9\}$
22	$\{6, 4\}$	45	$\{9, 6\}$	68	$\{11, 10\}$
23	$\{6, 5\}$	46	$\{9, 7\}$	69	$\{11, 11\}$

Table 4: Backward comparisons that obstruct a linear order.

n	x	exceptions with previous n_a values
14	$\{5, 1\}$	$n_a = 13, a = \{4, 4\}$: $C(x \rightarrow a) = 0.844794, C(a \rightarrow x) = 0.861353$
19	$\{6, 1\}$	$n_a = 16, a = \{5, 3\}$: $C(x \rightarrow a) = 0.897932, C(a \rightarrow x) = 0.898244$ $n_a = 17, a = \{5, 4\}$: $C(x \rightarrow a) = 0.857550, C(a \rightarrow x) = 0.898244$ $n_a = 18, a = \{5, 5\}$: $C(x \rightarrow a) = 0.825066, C(a \rightarrow x) = 0.898244$
25	$\{7, 1\}$	$n_a = 21, a = \{6, 3\}$: $C(x \rightarrow a) = 0.902627, C(a \rightarrow x) = 0.920782$ $n_a = 22, a = \{6, 4\}$: $C(x \rightarrow a) = 0.866396, C(a \rightarrow x) = 0.920782$ $n_a = 23, a = \{6, 5\}$: $C(x \rightarrow a) = 0.837432, C(a \rightarrow x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x \rightarrow a) = 0.813202, C(a \rightarrow x) = 0.920782$
26	$\{7, 2\}$	$n_a = 23, a = \{6, 5\}$: $C(x \rightarrow a) = 0.906054, C(a \rightarrow x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x \rightarrow a) = 0.878094, C(a \rightarrow x) = 0.920782$
32	$\{8, 1\}$	$n_a = 24, a = \{6, 6\}$: $C(x \rightarrow a) = 0.844632, C(a \rightarrow x) = 0.861654$ $n_a = 27, a = \{7, 3\}$: $C(x \rightarrow a) = 0.906390, C(a \rightarrow x) = 0.935785$ $n_a = 28, a = \{7, 4\}$: $C(x \rightarrow a) = 0.873076, C(a \rightarrow x) = 0.935785$ $n_a = 29, a = \{7, 5\}$: $C(x \rightarrow a) = 0.846552, C(a \rightarrow x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x \rightarrow a) = 0.824430, C(a \rightarrow x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x \rightarrow a) = 0.805397, C(a \rightarrow x) = 0.935785$
33	$\{8, 2\}$	$n_a = 29, a = \{7, 5\}$: $C(x \rightarrow a) = 0.911301, C(a \rightarrow x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x \rightarrow a) = 0.886261, C(a \rightarrow x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x \rightarrow a) = 0.864389, C(a \rightarrow x) = 0.935785$

continued on the next page

Table 4 continued from the previous page

n	x	exceptions with previous n_a values
34	$\{8, 3\}$	$n_a = 30, a = \{7, 6\}$: $C(x \rightarrow a) = 0.930247, C(a \rightarrow x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x \rightarrow a) = 0.906390, C(a \rightarrow x) = 0.935785$
40	$\{9, 1\}$	$n_a = 29, a = \{7, 5\}$: $C(x \rightarrow a) = 0.871798, C(a \rightarrow x) = 0.885622$ $n_a = 30, a = \{7, 6\}$: $C(x \rightarrow a) = 0.850314, C(a \rightarrow x) = 0.885622$ $n_a = 31, a = \{7, 7\}$: $C(x \rightarrow a) = 0.831776, C(a \rightarrow x) = 0.885622$ $n_a = 34, a = \{8, 3\}$: $C(x \rightarrow a) = 0.909518, C(a \rightarrow x) = 0.946395$ $n_a = 35, a = \{8, 4\}$: $C(x \rightarrow a) = 0.878388, C(a \rightarrow x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x \rightarrow a) = 0.853673, C(a \rightarrow x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x \rightarrow a) = 0.833105, C(a \rightarrow x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.815439, C(a \rightarrow x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.799923, C(a \rightarrow x) = 0.946395$
41	$\{9, 2\}$	$n_a = 35, a = \{8, 4\}$: $C(x \rightarrow a) = 0.942478, C(a \rightarrow x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x \rightarrow a) = 0.915285, C(a \rightarrow x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x \rightarrow a) = 0.892295, C(a \rightarrow x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.872314, C(a \rightarrow x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.854578, C(a \rightarrow x) = 0.946395$
42	$\{9, 3\}$	$n_a = 37, a = \{8, 6\}$: $C(x \rightarrow a) = 0.933809, C(a \rightarrow x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.912294, C(a \rightarrow x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.892960, C(a \rightarrow x) = 0.946395$
43	$\{9, 4\}$	$n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.944850, C(a \rightarrow x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.924272, C(a \rightarrow x) = 0.946395$

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Table 4 continued from the previous page

n	x	exceptions with previous n_a values
49	$\{10, 1\}$	$n_a = 35, a = \{8, 4\}$: $C(x \rightarrow a) = 0.899143, C(a \rightarrow x) = 0.903090$ $n_a = 36, a = \{8, 5\}$: $C(x \rightarrow a) = 0.875102, C(a \rightarrow x) = 0.903090$ $n_a = 37, a = \{8, 6\}$: $C(x \rightarrow a) = 0.855010, C(a \rightarrow x) = 0.903090$ $n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.837714, C(a \rightarrow x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.822502, C(a \rightarrow x) = 0.903090$ $n_a = 41, a = \{9, 2\}$: $C(x \rightarrow a) = 0.952333, C(a \rightarrow x) = 0.954243$ $n_a = 42, a = \{9, 3\}$: $C(x \rightarrow a) = 0.912182, C(a \rightarrow x) = 0.954243$ $n_a = 43, a = \{9, 4\}$: $C(x \rightarrow a) = 0.882762, C(a \rightarrow x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x \rightarrow a) = 0.859454, C(a \rightarrow x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x \rightarrow a) = 0.840087, C(a \rightarrow x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.823475, C(a \rightarrow x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.808903, C(a \rightarrow x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.795900, C(a \rightarrow x) = 0.954243$
50	$\{10, 2\}$	$n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.894640, C(a \rightarrow x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.877498, C(a \rightarrow x) = 0.903090$ $n_a = 43, a = \{9, 4\}$: $C(x \rightarrow a) = 0.943960, C(a \rightarrow x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x \rightarrow a) = 0.918479, C(a \rightarrow x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x \rightarrow a) = 0.897026, C(a \rightarrow x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.878445, C(a \rightarrow x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.862002, C(a \rightarrow x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.847215, C(a \rightarrow x) = 0.954243$

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Table 4 continued from the previous page

n	x	exceptions with previous n_a values
51	$\{10, 3\}$	$n_a = 45, a = \{9, 6\}$: $C(x \rightarrow a) = 0.936518, C(a \rightarrow x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.916673, C(a \rightarrow x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.898942, C(a \rightarrow x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.882860, C(a \rightarrow x) = 0.954243$
52	$\{10, 4\}$	$n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.947468, C(a \rightarrow x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.928787, C(a \rightarrow x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.911656, C(a \rightarrow x) = 0.954243$
53	$\{10, 5\}$	$n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.936626, C(a \rightarrow x) = 0.954243$

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Table 4 continued from the previous page

n	x	exceptions with previous n_a values
59	$\{11, 1\}$	$n_a = 38, a = \{8, 7\}$: $C(x \rightarrow a) = 0.856507, C(a \rightarrow x) = 0.867194$ $n_a = 39, a = \{8, 8\}$: $C(x \rightarrow a) = 0.841648, C(a \rightarrow x) = 0.867194$ $n_a = 43, a = \{9, 4\}$: $C(x \rightarrow a) = 0.900788, C(a \rightarrow x) = 0.916314$ $n_a = 44, a = \{9, 5\}$: $C(x \rightarrow a) = 0.878004, C(a \rightarrow x) = 0.916314$ $n_a = 45, a = \{9, 6\}$: $C(x \rightarrow a) = 0.859006, C(a \rightarrow x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.842681, C(a \rightarrow x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.828343, C(a \rightarrow x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.815542, C(a \rightarrow x) = 0.916314$ $n_a = 50, a = \{10, 2\}$: $C(x \rightarrow a) = 0.952868, C(a \rightarrow x) = 0.960253$ $n_a = 51, a = \{10, 3\}$: $C(x \rightarrow a) = 0.914492, C(a \rightarrow x) = 0.960253$ $n_a = 52, a = \{10, 4\}$: $C(x \rightarrow a) = 0.886455, C(a \rightarrow x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x \rightarrow a) = 0.864279, C(a \rightarrow x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x \rightarrow a) = 0.845875, C(a \rightarrow x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x \rightarrow a) = 0.830106, C(a \rightarrow x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x \rightarrow a) = 0.816285, C(a \rightarrow x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x \rightarrow a) = 0.803964, C(a \rightarrow x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.792835, C(a \rightarrow x) = 0.960253$

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Table 4 continued from the previous page

n	x	exceptions with previous n_a values
60	$\{11, 2\}$	$n_a = 45, a = \{9, 6\}$: $C(x \rightarrow a) = 0.915642, C(a \rightarrow x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x \rightarrow a) = 0.897629, C(a \rightarrow x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x \rightarrow a) = 0.881636, C(a \rightarrow x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.867225, C(a \rightarrow x) = 0.916314$ $n_a = 52, a = \{10, 4\}$: $C(x \rightarrow a) = 0.945256, C(a \rightarrow x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x \rightarrow a) = 0.921131, C(a \rightarrow x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x \rightarrow a) = 0.900885, C(a \rightarrow x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x \rightarrow a) = 0.883391, C(a \rightarrow x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x \rightarrow a) = 0.867946, C(a \rightarrow x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x \rightarrow a) = 0.854085, C(a \rightarrow x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.841485, C(a \rightarrow x) = 0.960253$
61	$\{11, 3\}$	$n_a = 48, a = \{9, 9\}$: $C(x \rightarrow a) = 0.902904, C(a \rightarrow x) = 0.916314$ $n_a = 53, a = \{10, 5\}$: $C(x \rightarrow a) = 0.959953, C(a \rightarrow x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x \rightarrow a) = 0.938697, C(a \rightarrow x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x \rightarrow a) = 0.920119, C(a \rightarrow x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x \rightarrow a) = 0.903587, C(a \rightarrow x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x \rightarrow a) = 0.888649, C(a \rightarrow x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.874982, C(a \rightarrow x) = 0.960253$

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Table 4 continued from the previous page

n	x	exceptions with previous n_a values
62	$\{11, 4\}$	$n_a = 55, a = \{10, 7\}$: $C(x \rightarrow a) = 0.949460, C(a \rightarrow x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x \rightarrow a) = 0.932147, C(a \rightarrow x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x \rightarrow a) = 0.916370, C(a \rightarrow x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.901830, C(a \rightarrow x) = 0.960253$
63	$\{11, 5\}$	$n_a = 56, a = \{10, 8\}$: $C(x \rightarrow a) = 0.956636, C(a \rightarrow x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x \rightarrow a) = 0.940214, C(a \rightarrow x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.924929, C(a \rightarrow x) = 0.960253$
64	$\{11, 6\}$	$n_a = 58, a = \{10, 10\}$: $C(x \rightarrow a) = 0.945667, C(a \rightarrow x) = 0.960253$

B.1 The global comparison clusters

On the infinite family of non-special integer signatures, we distinguish a non-strict and a strict comparison arrow:

$$a \longrightarrow b \iff C(a \rightarrow b) \geq C(b \rightarrow a), \quad a \xrightarrow{s} b \iff C(a \rightarrow b) > C(b \rightarrow a).$$

An arrow points from the greater element to the smaller one; equivalently, $a \xrightarrow{s} b$ means $b \prec a$. Equality produces two opposite non-strict arrows and no strict arrow. Define the corresponding clusters as the strongly connected components

$$\text{Cl}_{\geq}(a) = \{b : a \longrightarrow^* b \text{ and } b \longrightarrow^* a\},$$

and define $\text{Cl}_{>}(a)$ analogously using only $\xrightarrow{s}$. Reversing every arrow leaves these components unchanged.

Finite truncations require care. A path between two vertices of $\mathcal{S}_B$ may leave the shell, so a component of the induced subgraph need not be the intersection of a global component with $\mathcal{S}_B$. The 69-term prefix therefore gives components that are too small.

Calculations on nested shells suggest the following conjecture:

$$\text{Cl}_{\geq}(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \{\{2, 1\}, \{2, 2\}\}.$$

The computed component contains 112 of the 129 signatures in $\mathcal{S}_{12}$ and 2498 of the 2560 signatures in $\mathcal{S}_{18}$. It does not stabilize over $12 \leq B \leq 18$: enlarging the shell supplies new connecting paths. These counts support the conjecture but do not give a finite description of the global component.

For the strict graph, put

$$\begin{aligned} \mathcal{E}_s = \{ & \{2, 1\}, \{2, 2\}, \{3, 1\}, \{3, 2\}, \{3, 3\}, \{4, 1\}, \{4, 2\}, \{4, 3\}, \\ & \{2, 1, 1\}, \{2, 2, 1\}, \{2, 2, 2\}, \{2, 1, 1, 1\} \}. \end{aligned}$$

We conjecture that

$$\text{Cl}_>(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \mathcal{E}_s.$$

In $\mathcal{S}_{20}$, the computed strict component contains 6719 of 6738 signatures. Its intersection with $\mathcal{S}_{19}$ is $\mathcal{S}_{19} \setminus \mathcal{E}_s$; the other seven omitted vertices lie on the new boundary $a_1 + 2|a| = 20$. The computation does not prove that later shells absorb every boundary vertex, or that no element of $\mathcal{E}_s$ is eventually absorbed.

Unlike the three-cycle of Section 8, the comparisons in the rest of this appendix are computational observations and are not proved here: a tolerance of $2 \cdot 10^{-6}$ means only that smaller differences were treated as ties, not that the larger ones are certified.

For reference, the following cycle was computed in $\mathcal{S}_{14}$ with tolerance $2 \cdot 10^{-6}$. A shortest strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$ is

$$\{9, 1\} \xrightarrow{s} \{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\} \xrightarrow{s} \{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}.$$

Table 5 lists its edge rates.

Table 5: Strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$.		
Strict edge $a \xrightarrow{s} b$	$C(a \rightarrow b)$	$C(b \rightarrow a)$
$\{9, 1\} \xrightarrow{s} \{3, 3, 3\}$	0.630929754	0.500000000
$\{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\}$	0.792481250	0.753940561
$\{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\}$	0.527921767	0.500000000
$\{8, 4\} \xrightarrow{s} \{10, 1\}$	0.903089987	0.899142666
$\{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}$	0.500000000	0.477121255

A shortest return path has two edges:

$$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\} \xrightarrow{s} \{9, 1\}.$$

Table 6: Strict return path from $\{3, 2, 2, 2\}$ to $\{9, 1\}$.

Strict edge $a \xrightarrow{s} b$	$C(a \rightarrow b)$	$C(b \rightarrow a)$
$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\}$	0.564574819	0.500000000
$\{7, 5\} \xrightarrow{s} \{9, 1\}$	0.885621875	0.871797909

B.2 A length-three strict cycle

Strict cycles also occur at fixed length. Among the 55 non-special length-three signatures with entries at most 6, exhaustive computation at tolerance $2 \cdot 10^{-6}$ finds none. Allowing entries up to 7 gives

$$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}.$$

Table 7 gives both directed rates on every edge.

Table 7: A strict three-cycle of length-three signatures.

Strict edge $a \xrightarrow{s} b$	$C(a \rightarrow b)$	$C(b \rightarrow a)$	difference
$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\}$	0.920782221	0.910910734	0.009871488
$\{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\}$	0.924704131	0.920782221	0.003921910
$\{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}$	0.975533650	0.959474358	0.016059292

C Certification of the three-cycle

The cycle of Section 8 is the sharpest claim in this paper, and its narrowest edge has a margin of $3 \cdot 10^{-4}$. This appendix proves it. Write

$$c = \frac{\log 2}{\log 3}, \quad \alpha = \frac{\log 5}{\log 6}, \quad \gamma = \frac{\log 3}{\log 6}, \quad \theta = \frac{\log 5}{\log 3},$$

and recall $f_1 = \{5, 3\}$, $f_2 = \{3, 1, 1\}$, $f_3 = \{6, 1\}$.

Upper bounds on a rate are free: $C(g \rightarrow f)$ is an infimum, so every single β certifies one, and so does the endpoint. Lower bounds are the work, because $C(g \rightarrow f) \geq \kappa$ asserts $Z_g(\beta) \geq Z_f(\beta)^\kappa$ for every β . The three needed lower bounds are the following.

Lemma 2. $5^\beta + 3^\beta \geq (6^\beta + 1)^\alpha$ for all $\beta \geq 0$; hence $C(f_1 \rightarrow f_3) = \alpha$.

Proof. Put $t = 6^\beta \geq 1$, so that $5^\beta = t^\alpha$ and $3^\beta = t^\gamma$. As $0 < \alpha < 1$, the function $x \mapsto x^\alpha$ is concave, so $(t+1)^\alpha \leq t^\alpha + \alpha t^{\alpha-1}$ and

$$(t+1)^\alpha - t^\alpha - t^\gamma \leq \alpha t^{\alpha-1} - t^\gamma = t^{\alpha-1}(\alpha - t^{\gamma-\alpha+1}).$$

Now $\gamma - \alpha + 1 = 1 - \log(5/3)/\log 6 > 0$, because $6 > 5/3$; hence $t^{\gamma-\alpha+1} \geq 1 > \alpha$ for $t \geq 1$, the bracket is negative, and the inequality follows. Letting $\beta \rightarrow \infty$ gives the reverse bound $C(f_1 \rightarrow f_3) \leq \alpha$. $\square$

Lemma 3. $6^\beta + 1 \geq (3^\beta + 2)^c$ for all $\beta \geq 0$; hence $C(f_3 \rightarrow f_2) = c$.

Proof. Put $t = 3^\beta \geq 1$, so that $2^\beta = t^c$ and $6^\beta = t^{1+c}$. We first claim $c \log(t+2) \leq \log(t+1)$ for $t \geq 1$. The two sides agree at $t = 1$, both being $\log 2$, since $c \log 3 = \log 2$; and $c/(t+2) < 1/(t+1)$ because $c < 1$ gives $c(t+1) < t+2$. So the right-hand side grows faster and the claim holds. Therefore $(t+2)^c \leq t+1 \leq t^{1+c} + 1$, the last step because $t^{1+c} \geq t$ for $t \geq 1$. Taking $\beta = 0$ gives the reverse bound $C(f_3 \rightarrow f_2) \leq c$. $\square$

Lemma 4. $3^\beta + 2 > (5^\beta + 3^\beta)^c$ for all $\beta \geq 0$; hence $C(f_2 \rightarrow f_1) > c$.

Proof. Put $s = 3^\beta \geq 1$, so that $5^\beta = s^\theta$. Raising to the power $1/c = \log 3 / \log 2 > 1$, the claim reads $(s + 2)^{1/c} > s^\theta + s$. Convexity of $x \mapsto x^{1/c}$ gives $(s + 2)^{1/c} \geq s^{1/c} + (2/c) s^{1/c-1}$, so after division by s it suffices that $A(s) + B(s) > 1$, where

$$A(s) = s^{1/c-1} - s^{\theta-1}, \quad B(s) = \frac{2}{c} s^{1/c-2}.$$

Here $1/c - 1 > \theta - 1 > 0$: the first inequality is $(\log 3)^2 > \log 2 \log 5$, which holds as $1.2069 > 1.1156$. Hence $A(1) = 0$ and

$$A'(s) = s^{\theta-2} \left[(1/c - 1) s^{1/c-\theta} - (\theta - 1) \right] > 0 \quad (s \geq 1),$$

since $1/c - \theta > 0$, so A increases; and B decreases because $1/c - 2 < 0$. Finally

$$A(12) = 1.1030 \dots > 1, \quad B(12) = 1.1301 \dots > 1.$$

For $1 \leq s \leq 12$ we get $A(s) + B(s) \geq B(s) \geq B(12) > 1$, and for $s \geq 12$ we get $A(s) + B(s) \geq A(s) \geq A(12) > 1$. $\square$

Proposition 4. $f_1 \prec f_2 \prec f_3 \prec f_1$.

Proof. For $f_1 \prec f_2$: at $\beta = 0$ the ratio defining $C(f_1 \rightarrow f_2)$ is $\log 2 / \log 3 = c$, so $C(f_1 \rightarrow f_2) \leq c$, while Lemma 4 gives $C(f_2 \rightarrow f_1) > c$.

For $f_2 \prec f_3$: letting $\beta \rightarrow \infty$ gives $C(f_2 \rightarrow f_3) \leq \log 3 / \log 6 = \gamma$, while Lemma 3 gives $C(f_3 \rightarrow f_2) = c$. These differ in the right direction because $\gamma < c$ is $(\log 3)^2 < \log 2 \log 6$, that is $1.2069 \dots < 1.2419 \dots$.

For $f_3 \prec f_1$: evaluating at the single point $\beta = 9/20$,

$$C(f_3 \rightarrow f_1) \leq \frac{\log(6^{9/20} + 1)}{\log(5^{9/20} + 3^{9/20})} = 0.8979388 \dots,$$

while Lemma 2 gives $C(f_1 \rightarrow f_3) = \alpha = 0.8982444 \dots$. $\square$

Only three numerical comparisons enter: $A(12)$ and $B(12)$ against 1, the two products of logarithms, and the single evaluation at $\beta = 9/20$. Each is a finite computation with a margin of at least $3 \cdot 10^{-4}$. The script `analysis/certify_cycle.py` rechecks all five in interval arithmetic, so the verdicts are rigorous rather than floating point.